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Experience

Python Developer **CloudTechTiq** Nov 2024 – Present

- Architected a multi-tenant hosting and billing/ERP SaaS on **Django REST Framework** spanning 25+ apps and 230+ ViewSets under a unified `/api/` namespace.
- Designed a dual-gateway payment system (**Razorpay** + **Stripe**) with HMAC-SHA256 webhook idempotency to prevent duplicate charges, coordinating payment, invoicing, provisioning, and double-entry ledger reconciliation, plus Indian GST e-invoicing (EINV-2.1) through the NIC gateway with IRN generation and signed QR codes.
- Extracted billing into a standalone **FastAPI** microservice (billing-ms) as the platform's single billing authority: double-entry ledger, multi-invoice payment allocation and refunds, proforma –> final invoice posting, and multi-entity issuance that resolves legal entity, currency and GST/VAT from the billing address.
- Built an OAuth2/OpenID authorization server and **FastAPI** gateway to **Apache CloudStack** for VMs, volumes, networks and templates, with webhook-first state sync, a polling fallback, and live job progress streamed to clients over **SSE**.
- Secured the platform with RS512 **JWT** auth (kid-keyed JWKS, refresh-token rotation over HTTPOnly cookies), a 2FA flow over email OTP and RFC-6238 TOTP, and a 43-permission department-scoped RBAC.
- Shipped real-time notifications over a **Django Channels** 4 + Redis WebSocket server with JWT-authenticated async consumers, backed by a 24-task **Celery** system (late-ack, worker-lost requeue).
- Built a service-provisioning engine that tracks each order through its lifecycle across seven service types, automating bare-metal setup over an HPE iLO Redfish wrapper and Celery-driven IP allocation.
- Built and open-sourced a type-safe Namecheap SSL/domain SDK (**httpx**, **Pydantic v2**, mypy-strict), published to PyPI — github.com/alzywelzy/namecheap-wrapper.
- Published open-source **Python** services: Shortly, a URL shortener (FastAPI, PostgreSQL, Redis read-through cache, Docker Compose, systemd) and fastapi-genesis, a feature-sliced FastAPI starter on async SQLAlchemy 2, Alembic, uv, ruff and ty.
- Planned service migrations and cutovers with written runbooks (a check and a rollback per step), architecture decision records, and a test-parity gate so every existing test is ported or explicitly dropped before release.
- Designed scheduled and queued background work with leases, heartbeats, retries and dead-letter queues so failed jobs surface instead of disappearing.
- Eliminated N+1 queries via Prefetch/OuterRef/Exists across a dual async/sync database layer (asyncpg + psycopg) on SQLAlchemy 2.0 with connection pooling and keyset cursor pagination.

Education

Bachelor of Computer Applications (BCA) **Bundelkhand University** Jul 2021 – Jun 2024

- Built an AI-powered news summarizer in Python as a final-year academic project, using the OpenAI API to condense full-length articles into short, readable digests.

Technical Skills

- **Languages:** Python, TypeScript, JavaScript, SQL, Bash
- **Backend:** Django, Django REST Framework, FastAPI, Hono, Celery, SQLAlchemy, Drizzle, Pydantic, Zod, WebSockets, SSE
- **Data:** PostgreSQL, Redis, MySQL, MongoDB, Cloudflare D1/R2, RabbitMQ, Alembic
- **DevOps & Cloud:** Docker, Docker Compose, Docker Swarm, Kubernetes, GitHub Actions, GHCR, NGINX, Gunicorn, Linux/systemd, AWS (EC2, S3, RDS, Lambda), GCP, Azure, Cloudflare (Workers, Pages, DNS), Apache CloudStack
- **Security & Payments:** OAuth2/OpenID, JWT/JWKS, WebAuthn/passkeys, 2FA/TOTP, RBAC, HMAC, Stripe, Razorpay
- **Observability & Testing:** OpenTelemetry, Prometheus, Grafana, Loki, Tempo; pytest, Vitest, Playwright, CI/CD
- **AI & Tooling:** OpenAI API, LangChain; uv, Poetry, ruff, mypy, Biome, ESLint